

Gujarat Hackathon 2026 – Build AI Agent using IBM Bob

Challenge 1 – Smart Salinity Ingress & Coastal Farmland Protection Advisor

Domain: Agriculture & Coastal Resilience

The Challenge

Coastal districts of Gujarat such as Bhavnagar, Jamnagar, and Kutch face increasing seawater ingress into groundwater and farmland, degrading soil quality and reducing crop yields. Farmers lack timely, localized guidance on salinity-resistant crops and mitigation practices.

Objective

Build Agentic AI solutions that monitor soil and groundwater salinity trends, recommend salt-tolerant crop alternatives, and guide farmers on land reclamation and irrigation practices.

Suggested AI Agents

- Soil & Groundwater Salinity Monitoring Agent
- Salinity-Resistant Crop Advisory Agent
- Irrigation & Land Reclamation Planning Agent
- Farmer Alert & Advisory Agent
- Coastal Farmland Health Dashboard Agent

Technology to be used – IBM Bob + IBM Granite LLM + IBM cloud

Challenge 2 – Smart Migrant Labor Welfare & Skill Mapping Platform

Domain: Social Governance

The Challenge

Gujarat hosts a large interstate migrant workforce in construction, textiles, and diamond polishing, many of whom lack access to welfare schemes, fair wage information, and safe working condition reporting, leaving them vulnerable to exploitation.

Objective

Develop Agentic AI solutions that map migrant labour skills and locations, connect them to welfare entitlements, and flag unsafe or exploitative working conditions.

Suggested AI Agents

- Migrant Skill & Location Mapping Agent
- Welfare Scheme Eligibility Agent
- Wage Fairness Monitoring Agent
- Grievance & Safety Reporting Agent
- Labor Welfare Dashboard Agent

Technology to be used – IBM Bob + IBM Granite LLM + IBM cloud

Challenge 3 – Smart Rann of Kutch Eco-Tourism & Carrying Capacity Planner

Domain: Tourism

The Challenge

The Rann Utsav and Kutch region attract heavy seasonal tourist inflow, straining fragile desert ecology, local water resources, and infrastructure, while visitors lack guidance on sustainable and lesser-known eco-tourism alternatives.

Objective

Build Agentic AI solutions that manage tourist load, recommend sustainable travel itineraries, and protect ecologically sensitive zones in Kutch and surrounding regions.

Suggested AI Agents

- Tourist Load Forecasting Agent
- Sustainable Itinerary Recommendation Agent
- Ecological Carrying Capacity Agent
- Local Artisan & Community Linkage Agent
- Tourism Impact Dashboard Agent

Technology to be used – IBM Bob + IBM Granite LLM + IBM cloud

Challenge 4 – AI-Based Gir Forest Human-Wildlife Conflict Mitigation System

Domain: Environmental Sustainability

The Challenge

Communities around Gir Forest face recurring human-wildlife conflict involving Asiatic lions and leopards straying into villages, leading to livestock loss, human injury, and retaliatory harm to wildlife due to delayed alerts and poor coordination.

Objective

Create Agentic AI solutions that predict wildlife movement near human settlements, issue real-time alerts, and coordinate rapid response between forest officials and villagers.

Suggested AI Agents

- Wildlife Movement Prediction Agent
- Village Alert & Early Warning Agent
- Forest Department Response Coordination Agent
- Livestock Loss Compensation Assistant Agent
- Conflict Hotspot Dashboard Agent

Technology to be used – IBM Bob + IBM Granite LLM + IBM cloud

Challenge 5 – Smart Narmada Canal Water Distribution & Equity Monitor

Domain: Water Resource Management

The Challenge

Unequal and inefficient distribution of Narmada canal water across head-reach and tail-end farmers in Gujarat leads to disputes, crop stress, and inefficient water usage, especially during peak irrigation seasons.

Objective

Develop Agentic AI solutions that monitor canal water flow, optimize equitable distribution schedules, and provide farmers with real-time water availability information.

Suggested AI Agents

- Canal Flow Monitoring Agent
- Equitable Distribution Scheduling Agent
- Farmer Water Availability Alert Agent
- Dispute Detection & Mediation Agent
- Irrigation Efficiency Dashboard Agent

Technology to be used – IBM Bob + IBM Granite LLM + IBM cloud

Challenge 6 – AI-Driven Cyclone & Coastal Disaster Early Warning System

Domain: Disaster Management

The Challenge

Gujarat's 1,600+ km coastline is highly vulnerable to cyclones (e.g., Tauktae, Biparjoy), yet coastal communities and fishermen often receive delayed or generalized warnings, leading to preventable loss of life, boats, and property.

Objective

Build Agentic AI solutions that provide hyperlocal cyclone tracking, fishermen safety alerts, and coordinated evacuation and relief planning for coastal districts.

Suggested AI Agents

- Cyclone Track & Intensity Prediction Agent
- Fishermen Safety Alert Agent
- Evacuation Route Planning Agent
- Relief Resource Coordination Agent
- Post-Disaster Damage Assessment Agent

Technology to be used – IBM Bob + IBM Granite LLM + IBM cloud

Challenge 7 – Smart Urban Flooding & Drainage Management System for Ahmedabad-Surat

Domain: Urban Infrastructure

The Challenge

Rapid urbanization in cities like Ahmedabad and Surat has outpaced stormwater drainage infrastructure, causing frequent monsoon flooding, traffic disruption, and property damage due to poor predictive planning and delayed civic response.

Objective

Create Agentic AI solutions that predict flood-prone zones, optimize drainage maintenance schedules, and coordinate real-time civic response during heavy rainfall events.

Suggested AI Agents

- Flood Risk Prediction Agent
- Drainage Maintenance Scheduling Agent
- Real-Time Civic Response Coordination Agent
- Citizen Flood Reporting Agent
- Urban Resilience Dashboard Agent
- Post-Disaster Damage Assessment Agent

Technology to be used – IBM Bob + IBM Granite LLM + IBM cloud

Challenge 8 – AI-Based Dairy Cooperative Supply Chain Optimization Platform

Domain: Economic Development

The Challenge

Gujarat's dairy cooperatives (Amul model) face challenges in milk collection route efficiency, fodder cost volatility, and matching milk supply with fluctuating demand, impacting profitability for millions of dairy farmers.

Objective

Develop Agentic AI solutions that optimize milk collection logistics, forecast fodder prices, and align production planning with cooperative and market demand.

Suggested AI Agents

- Milk Collection Route Optimization Agent
- Fodder Price Forecasting Agent
- Demand-Supply Matching Agent
- Farmer Payment & Transparency Agent
- Cooperative Performance Dashboard Agent

Technology to be used – IBM Bob + IBM Granite LLM + IBM cloud

Challenge 9 – Smart Industrial Pollution Monitoring for Golden Corridor (Vapi–Ankleswar)

Domain: Environmental Sustainability

The Challenge

The industrial belt along Vapi, Ankleshwar, and Vatva faces significant air and water pollution from chemical and manufacturing units, with limited real-time monitoring and slow regulatory response to violations.

Objective

Build Agentic AI solutions that continuously monitor industrial emissions and effluents, detect violations, and trigger timely regulatory and community alerts.

Suggested AI Agents

- Emission & Effluent Monitoring Agent
- Violation Detection & Compliance Agent
- Regulatory Alert & Escalation Agent
- Public Health Risk Assessment Agent
- Industrial Pollution Dashboard Agent

Technology to be used – IBM Bob + IBM Granite LLM + IBM cloud

Challenge 10 – Intelligent Drought & Groundwater Depletion Advisor for Saurashtra

Domain: Drought & Water Management

The Challenge

Saurashtra region experiences recurring drought conditions and severe groundwater depletion due to over-extraction for agriculture, with farmers lacking timely guidance on water-efficient cropping and recharge practices.

Objective

Create Agentic AI solutions that assess groundwater depletion trends, recommend water-efficient farming practices, and guide community-level recharge interventions.

Suggested AI Agents

- Groundwater Depletion Monitoring Agent
- Water-Efficient Crop Advisory Agent
- Recharge Structure Planning Agent
- Drought Early Warning Agent
- Community Water Dashboard Agent

Technology to be used – IBM Bob + IBM Granite LLM + IBM cloud

Challenge 11 – AI-Based Salt Pan Worker (Agariya) Welfare & Safety Platform

Domain: Social Governance

The Challenge

Salt pan workers (Agariyas) in the Little Rann of Kutch work in extreme, isolated conditions with limited access to healthcare, fair pricing for salt produce, and welfare scheme benefits, leading to chronic health and economic vulnerability.

Objective

Develop Agentic AI solutions that connect Agariya communities to healthcare outreach, fair market pricing information, and applicable welfare schemes.

Suggested AI Agents

- Remote Healthcare Outreach Agent
- Salt Price Discovery Agent
- Welfare Scheme Matching Agent
- Worker Safety Monitoring Agent
- Community Support Dashboard Agent

Technology to be used – IBM Bob + IBM Granite LLM + IBM cloud

Challenge 12: Smart Heritage Conservation Platform for Ahmedabad's Walled City & Modhera

Domain: Tourism & Heritage

The Challenge

Gujarat's UNESCO heritage sites, including Ahmedabad's Walled City and Modhera Sun Temple, face conservation risks from urban encroachment, weathering, and unmanaged tourist footfall, alongside underutilized digital storytelling for visitors.

Objective

Build Agentic AI solutions that monitor heritage site structural health, manage visitor flow, and deliver immersive, personalized cultural storytelling experiences.

Suggested AI Agents

- Structural Health Monitoring Agent
- Visitor Flow Management Agent
- Personalized Heritage Storytelling Agent
- Encroachment Detection Agent
- Conservation Reporting Dashboard Agent

Technology to be used – IBM Bob + IBM Granite LLM + IBM cloud

Challenge 13 – AI-Powered Cotton & Groundnut Market Linkage Platform

Domain: Economic Development

The Challenge

Gujarat is a leading producer of cotton and groundnut, yet farmers frequently face price exploitation by middlemen, lack access to real-time mandi prices, and have limited bargaining power in direct market linkages.

Objective

Create Agentic AI solutions that provide real-time price intelligence, connect farmers directly with buyers, and support informed selling and storage decisions.

Suggested AI Agents

- Mandi Price Forecasting Agent
- Direct Buyer-Farmer Matching Agent
- Storage & Selling Timing Advisor Agent
- Quality Grading Assistance Agent
- Farmer Income Dashboard Agent

Technology to be used – IBM Bob + IBM Granite LLM + IBM cloud

Challenge 14 – Smart Renewable Energy (Solar-Wind Hybrid) Asset Monitoring for Kutch & Banaskantha

Domain: Energy & Sustainability

The Challenge

Gujarat leads India in hybrid solar-wind renewable capacity, particularly in Kutch and Banaskantha, but scattered installations face inefficiencies from inconsistent maintenance scheduling, underperformance detection, and grid integration challenges.

Objective

Develop Agentic AI solutions that monitor renewable asset performance, predict maintenance needs, and optimize grid integration for hybrid energy parks.

Suggested AI Agents

- Asset Performance Monitoring Agent
- Predictive Maintenance Agent
- Grid Integration Optimization Agent
- Weather-Based Generation Forecasting Agent
- Renewable Energy Dashboard Agent

Technology to be used – IBM Bob + IBM Granite LLM + IBM cloud

Challenge 15 – Smart Energy Management & Grid Balancing Platform

Domain: Energy & Sustainability

The Challenge

Gujarat's growing solar capacity and fluctuating electricity demand create challenges in balancing grid load, managing battery storage, and predicting outages, leading to inefficiencies and reliability concerns for consumers and industries.

Objective

Build Agentic AI solutions where smart grid agents balance electricity demand, solar generation, battery storage, and outage prediction to ensure stable and efficient power delivery.

Suggested AI Agents

- Demand-Supply Balancing Agent
- Solar Generation Forecasting Agent
- Battery Storage Optimization Agent
- Outage Prediction Agent
- Grid Performance Dashboard Agent

Technology to be used – IBM Bob + IBM Granite LLM + IBM cloud

Challenge 16 – Fisheries & Coastal Livelihood Intelligence Platform

Domain: Economic Development

The Challenge

Gujarat has India's longest coastline, yet fishermen contend with unpredictable marine weather, fluctuating fish availability, and volatile market prices, which together threaten their safety and income stability.

Objective

Develop Agentic AI solutions that deliver marine weather alerts, identify promising fishing zones, optimize fuel consumption, and forecast fish market demand for coastal communities.

Suggested AI Agents

- Marine Weather Agent
- Fishing Zone Prediction Agent
- Market Price Agent
- Fuel Optimization Agent
- Safety Alert Agent

Technology to be used – IBM Bob + IBM Granite LLM + IBM cloud

Challenge 17 – Intelligent Port Logistics & Cargo Coordination Platform

Domain: Logistics

The Challenge

Gujarat is home to some of India's busiest ports, where efficient cargo movement, berth allocation, customs coordination, and logistics planning remain persistent operational bottlenecks affecting trade efficiency.

Objective

Create Agentic AI solutions that optimize cargo scheduling, predict port congestion, coordinate logistics stakeholders, and reduce vessel and truck turnaround time.

Suggested AI Agents

- Vessel Scheduling Agent
- Cargo Planning Agent
- Customs Documentation Agent
- Truck Routing Agent
- Port Operations Dashboard Agent

Technology to be used – IBM Bob + IBM Granite LLM + IBM cloud

Challenge 18: Municipal Solid Waste & Circular Economy Agent

Domain: Urban Waste Management

The Challenge

Rapid urbanization across Gujarat's municipal corporations has outpaced door-to-door segregation compliance and collection-route efficiency in many wards, creating a gap between mandatory segregation policies and ground-level implementation.

Objective

Create Agentic AI solutions that optimize waste-collection routes dynamically, improve segregation compliance through citizen nudges, automatically route citizen grievances to the appropriate municipal desk, and provide ward-level analytics for better decision-making.

Suggested AI Agents

- Route Optimization Agent
- Segregation Compliance Tracking Agent
- Grievance Intake Agent (Multilingual Chat/Voice)
- Municipal Routing Agent
- Ward Analytics Dashboard Agent

Technology to be used – IBM Bob + IBM Granite LLM + IBM Cloud

Challenge 19: Rural & Tribal Healthcare Access Agent

Domain: Rural Healthcare

The Challenge

Tribal belts like the Dangs, Narmada, and Chhota Udepur districts face persistent gaps in healthcare access due to long distances to facilities, shortages of specialists, and ASHA workers being stretched thin across large areas.

Objective

Create Agentic AI solutions that perform preliminary symptom triage through Gujarati and tribal-dialect-aware voice input, connect patients to teleconsultation when appropriate, and assist ASHA and PHC workers in tracking medicine stock and follow-ups for chronic patients.

Suggested AI Agents

- Voice Triage Agent
- Severity Classification Agent
- Teleconsult Scheduling Agent
- Medicine Stock & Follow-up Tracking Agent

Technology to be used – IBM Bob + IBM Granite LLM + IBM Cloud

Challenge 20 – AI-Powered Personalized Learning & Teacher Assistant Platform

Domain: Education

The Challenge

Many government and rural schools in Gujarat face challenges such as diverse student learning levels, limited access to personalized learning resources, increasing administrative workload for teachers, and insufficient real-time insights into student progress. Teachers require AI-powered assistance to create engaging lesson plans, assess student performance, and provide individualized learning support while reducing repetitive tasks.

Objective

Create Agentic AI solutions that personalize student learning experiences, assist teachers in lesson planning and content generation, automate assessment and feedback, identify students requiring additional support, and provide actionable insights to improve learning outcomes.

Suggested AI Agents

- Personalized Learning Recommendation Agent
- AI Lesson Plan & Content Generation Agent
- Student Performance & Learning Analytics Agent
- Automated Assessment & Feedback Agent
- Teacher Assistant & Classroom Support Agent
- IBM SkillsBuild learning resources
- Open Educational Resources (OER)

Technology to be used - IBM Bob + IBM Granite LLM + IBM Cloud